

Effects of four commercial fungal formulations on mortality and sporulation in house flies (*Musca domestica*) and stable flies (*Stomoxys calcitrans*)

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Abstract. The house fly *Musca domestica* L. (Diptera: Muscidae) and stable fly *Stomoxys calcitrans* (L.) (Diptera: Muscidae) are major pests of livestock. Biological control is an important tool in an integrated control framework. Increased mortality in filth flies has been documented with entomopathogenic fungi, several strains of which are commercially available. Three strains of *Beauveria bassiana* (Balsamo-Crivelli) Vuillemin (Hypocreales: Cordycipitaceae) and one strain of *Metarhizium brunneum* (Petch) (Hypocreales: Clavicipitaceae) were tested in commercial formulations for pathogenicity against house flies and stable flies. There was a significant increase in mortality of house flies with three of the formulations, BotaniGard[®] ES, Mycotrol[®] O, and Met52[®] EC, during days 4–9 in comparison with balEnce[™] and the control. In stable flies, mortality rates were highest with Met52[®] EC, followed by Mycotrol[®] O, BotaniGard[®] ES and, finally, balEnce[™]. There was a significant fungal effect on sporulation in both house flies and stable flies. Product formulation, species differences and fungal strains may be responsible for some of the differences observed. Future testing in field situations is necessary. These commercial biopesticides may represent important tools in integrated fly management programmes.

Key words. *Beauveria bassiana*, *Metarhizium anisopliae*, *Metarhizium brunneum*, biological control, entomopathogenic, organic.

Introduction

The house fly, *Musca domestica* L., and stable fly, *Stomoxys calcitrans* (L.), are major medical and veterinary pests in both urban and rural environments. These flies can transmit harmful pathogens to humans and animals. Both species cause irritation and annoyance to employees on farms, reduce animal welfare, and cause declines in the productivity of livestock and poultry (Malik *et al.*, 2007; Mishra *et al.*, 2011; Taylor *et al.*, 2012). In addition, as both species breed in organic matter, they

are problematic in areas in which organic waste is stored, particularly in waste management facilities.

Pest management of these flies has relied historically on the use of insecticides (Scott & Rutz, 1988). However, flies have developed resistance to many of the available active ingredients (Keiding, 1999; Scott *et al.*, 2000, 2013; Kaufman *et al.*, 2010). Moreover, many of the insecticides used to manage flies also have deleterious effects on non-target organisms, including those that are biological control agents, such as predators and parasitoids (Scott *et al.*, 1991; Geden *et al.*, 1992).

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An integrated programme for pest management involving multiple strategies, including cultural (waste management), physical (screens), mechanical (traps and vacuums) and biological (pathogens, predators and parasites) control methods, in addition to chemical control using insecticides, is more likely to provide longterm control in a more cost-effective and sustainable way. Biological control is an important part of an integrated pest management programme for fly control. Entomopathogenic fungi are naturally occurring biological control agents that can cause filth fly mortality (Steinkraus *et al.*, 1990; Skovgaard & Steenberg, 2002). The entomopathogenic fungi *Beauveria bassiana* and *Metarhizium anisopliae sensu lato* have been investigated for use in the control of house flies and stable flies, with successful outcomes in some cases (Watson *et al.*, 1995; Kaufman *et al.*, 2005, 2008; Anderson *et al.*, 2011; Mishra *et al.*, 2011).

A substantial number of commercial mycopesticides have been developed, in the majority of which *B. bassiana* and *M. anisopliae s.l.* serve as active ingredients (de Faria & Wraight, 2007). Manipulation of the fungal strains and quantity naturally present in the environment, and the development of a convenient method of application, may accelerate fly mortality and assist in house fly management. The major challenge associated with the adoption of this technology concerns the creation of formulations. The persistence of *B. bassiana* products has been demonstrated to be reduced in the field compared with that observed during laboratory studies (Acharya *et al.*, 2015a, 2015b). Conidia remained viable for up to 13 weeks in laboratory conditions, but for only 1–2 weeks in the field (Acharya *et al.*, 2015a). An appropriate formulation is needed to combat the hydrophobic qualities of conidia, to ensure an even distribution of conidia throughout the treated area, and to improve the persistence of conidia exposed to ultraviolet light, high temperatures and other detrimental environmental conditions (Morley-Davis *et al.*, 1996), while maintaining pathogenicity.

The effective use of fungi as biopesticides is dependent on strain, dosage, application and formulation. Few of these fungal strains and formulations have been evaluated for pathogenicity in house flies and stable flies in the forms in which they are commercially available. Furthermore, laboratory evaluations typically involve forced-contact exposure methods, such as the immersion or dipping of the organism in the formulation. Such methods are effective but unnatural and are likely to result in much higher mortality than would occur following a natural exposure. Thus, the purpose of this study was to test the efficacy of four commercial fungal formulations containing *B. bassiana* and *Metarhizium brunneum* when applied in a natural exposure system for the potential control of house flies and stable flies. Rates of fly mortality and sporulation of cadavers were compared among the commercial formulations and a control treatment.

Materials and methods

House and stable flies

The house fly strain used has been in culture for > 50 years without exposure to pesticides or entomopathogenic fungi. These flies were reared at the University of Florida under

laboratory conditions of 26 °C, 45–55% relative humidity (RH) and a photoperiod of LD 16:8 h. Larval diet consisted of calf protein supplement, wood chips, bran and tap water at a ratio of 2:3:15:8, respectively, and adults were maintained on water, non-fat dried milk and sucrose (Kaufman *et al.*, 2001).

Stable flies were sourced from the U.S. Department of Agriculture Agricultural Research Service (USDA-ARS), Center for Medical, Agricultural and Veterinary Entomology (CMAVE) (Gainesville, FL, U.S.A.) colony. This colony has been maintained for ~35 years at 26 ± 2 °C, 60 ± 5% RH and a photoperiod of LD 12:12 h. Adult stable flies were provided with citrated bovine blood *ad libitum*. Larval diet was as described in Machtinger *et al.* (2014). Flies of both species used in experiments were 3–5 days old. Flies were anaesthetized using carbon dioxide (CO₂) and sorted by sex prior to use in bioassays.

Fungal strains

Commercial strains of entomopathogenic fungi were purchased for pathogenicity testing; these included two strains of *B. bassiana* and one strain of *M. brunneum*. The *B. bassiana* strains tested were HF23, a house fly-derived strain in the formulations labelled balEnce™ (Terregena, Inc., Raleigh, NC, U.S.A.), and GHA (Laverlam International Corp., Butte, MT, U.S.A.), a strain isolated from a *Diabrotica* sp. (Coleoptera: Chrysomelidae) in two different formulations, BotaniGard® ES (BioWorks, Inc., Victor, NY, U.S.A.) and Mycotrol® O (BioWorks, Inc.). The percentage of active ingredient varied with the product and amounted to 11.3%, 10.9% and 1.29% conidia in BotaniGard® ES, Mycotrol® O and balEnce™, respectively. The *M. brunneum* strain tested was F52, a strain isolated from *Cydia pomonella* (L.) (Lepidoptera: Tortricidae), commercially available as Met52® EC with 11% active ingredient (Novozymes Biologicals, Inc., Franklinton, NC, U.S.A.). All of the formulations were emulsifiable suspensions that used petroleum distillates as emulsifying agents, except for Mycotrol® O, which contains vegetable oils and is labelled as suitable for organic use.

The viability of fungal cultures was assessed by germinating propagules on agar as follows. Commercial solutions were spread on Sabouraud's dextrose agar with yeast (SDAY; yeast 0.5%) and water agar in a concentration gradient to ensure that accurate observation of the conidia was possible. Plates were incubated at 22.5 °C and 55% RH for 18–24 h and the percentages of conidia that germinated were calculated for each strain. All formulations allowed a germination rate of > 90%, except balEnce™, which showed a germination rate of < 20%. Prior to inoculation, the concentration of conidia in each commercial solution was calculated using a haemocytometer. Solutions were then calibrated by dilution in deionized water with Tween 80 (0.2%) and horticultural oil (SunSpray® Ultra-Fine® Spray Oil; HollyFrontier® Refining and Marketing LLC, Plymouth Meeting, PA, U.S.A.), taking into consideration both the quality (percentage viability) and quantity (count) of conidia. As the formulations were oil-based, horticultural oil was added so that the amount of oil was equalized across all solutions (~1.8%). As a control, horticultural oil was added to deionized water (and Tween 80) and applied to filter papers (0.25% concentration).

Bioassays

Filter papers (P5; 9.0 cm in diameter; Fisher Scientific, Inc., Pittsburgh, PA, U.S.A.) were placed in the lids of 100 × 15-mm polystyrene Petri dishes (Fisher Scientific, Inc.). One-millilitre solutions of balEnce™, BotaniGard® ES, Mycotrol® O and Met52® EC (1×10^8 conidia/mL) in deionized water were applied to the filter papers so that the whole area was covered, resulting in a concentration of 1.57×10^6 conidia/cm². Filter papers were left for 2 h to dry in a laminar flow hood (*c.* 22.5 °C). Five flies, separated by sex, were added to each dish; 25 female and 25 male flies were tested in each treatment for each replicate (10 Petri dishes per treatment per replicate). Anaesthetized flies were placed in the untreated base of the Petri dish and the lid with the treated filter paper was placed on top. The dishes were left in this position to allow the flies to recover from the anaesthesia. After 10 min, the dishes were flipped over and then left for 60 min in the laminar flow hood. After exposure, flies were anaesthetized with CO₂, removed from the dishes and placed into holding containers in their exposure groups. Mortality was monitored after each anaesthetic event.

House flies were held in 188.3-mL plastic soufflé cups with plastic lids (Plastic Container City, Brooklyn, NY, U.S.A.). The base of each cup was covered with a filter paper disc (P5; 5.5 cm in diameter; Fisher Scientific, Inc.) to absorb excess moisture. Lids were perforated with a cross to allow the insertion of a dental wick and the provision of a 10% sugar solution. Stable flies were held in 354.9-mL plastic cups (Plastic Container City). The bottom of the cup was fitted with mesh screen (#12 mesh), and cups were held upside-down with a filter paper (P5; 5.5 cm in diameter; Fisher Scientific, Inc.) at the base. Stable flies were provided with a 10% sugar solution in a 29.6-mL plastic cup (Plastic Container City), the lid of which was perforated to allow the insertion of a dental wick. Citrated bovine blood was provided to stable flies every 36–48 h in a 14.8-mL cup (Plastic Container City) using a small piece of cotton ball and screening (#12 mesh) to prevent drowning.

Cups with house flies were held within 7.7-L plastic boxes measuring 30.5 × 22.8 × 10.2 cm (Pioneer Plastics, Inc., Dixon, KY, U.S.A.) and cups with stable flies were held in 70-L plastic storage bins (Sterilite Corp., Townsend, MA, U.S.A.) to maintain humidity. Each species was isolated by treatment and held at room environmental conditions of *c.* 22.5 °C and 55% RH. Dead flies were removed from the containers every day until all flies had died. Cadavers were washed in 95% ethanol, dried and placed into individual wells of a 96-well plate (Fisher Scientific, Inc.). Plates were covered with moistened filter paper and held for an additional 14 days and inspected for sporulation every 7 days. Flies were considered to have sporulated if external growth of the fungus was visible on the cadaver using a stereomicroscope at ×15 magnification. The trial was replicated four times for house flies and five times for stable flies.

Statistical analysis

Mortality was analysed for each species individually by fitting linear mixed models that considered the repeated nature

of the data and consisted of fixed-effect factors of trial (as a blocking factor), time of measurement (0–20 days), sex (male, female) and treatment (control, Met52® EC, Mycotrol® O, BotaniGard® ES, balEnce™), together with all two- and three-way interactions among time, sex and treatment. In addition, residual terms were modelled by considering an autoregressive order 1 error structure. Least significant difference (LSD) tests with Bonferroni correction were obtained to compare mean mortalities among treatments and across time ($\alpha = 0.05$). Based on the same models, lethal time at 50% and 90% (i.e. LT₅₀ and LT₉₀) were obtained for each species and treatment × sex combination. All analyses were conducted in SAS Version 9.3 (SAS, 2008).

Sporulation was analysed for each species individually by fitting a generalized linear model for a binomial distribution with a logit link. The model consisted of the fixed effects of trial (for blocking) and treatment (control, Met52® EC, Mycotrol® O, BotaniGard® ES, balEnce™). For house flies, it was necessary to correct for over-dispersion. Least significant difference tests were used to compare sporulation (proportion) between the control and fungal treatments ($\alpha = 0.05$). All analyses were conducted in SAS Version 9.3 (SAS, 2008).

Results

Mortality

All factors (i.e. time, sex and treatment) and tested interactions had significant effects on mortality in the house fly ($P < 0.0001$), including the three-way interaction of time, sex and treatment. In relation to sex, the fungal product had a significant effect on mortality, which occurred earlier (3–9 days; data not shown) in male house flies than in female house flies (4–12 days; data not shown). The interaction of treatment over time was also significant ($F_{80,3838} = 4.65$, $P < 0.0001$). Three of the products, BotaniGard® ES, Mycotrol® O and Met52® EC, produced significant increases in mortality during days 4–9 compared with balEnce™ and the control. Significant differences across time between the fungal treatment balEnce™ and the control were rare (Fig. 1). All the house flies were dead in the Mycotrol® O treatment by day 7, in the Met52® EC and BotaniGard® ES treatments by day 10, and in the balEnce™ treatment by day 13. In the control treatment, some individuals survived until day 19. House fly LT₅₀ values showed that males reached 50% mortality faster than females in all treatments including the control (Fig. 2). The two GHA strain treatments (BotaniGard® ES and Mycotrol® O) achieved LT₅₀ and LT₉₀ values faster than the other treatments.

Of the factors (i.e. time, sex and treatment) and interactions tested, only sex ($F_{1,621} = 13.82$, $P = 0.0002$), treatment ($F_{1,621} = 13.82$, $P < 0.0001$) and the interaction between time and treatment ($F_{84,5092} = 4.74$, $P < 0.0001$) had significant effects on mortality in the stable fly. The biological relevance of the significant effect of sex on mortality irrespective of time and treatment was slight. The proportion of mortality was 0.68 ± 0.005 in females and 0.70 ± 0.005 in males. There were significant increases in mortality with all of the products from day 5 to day 14 compared with the control (Fig. 3).

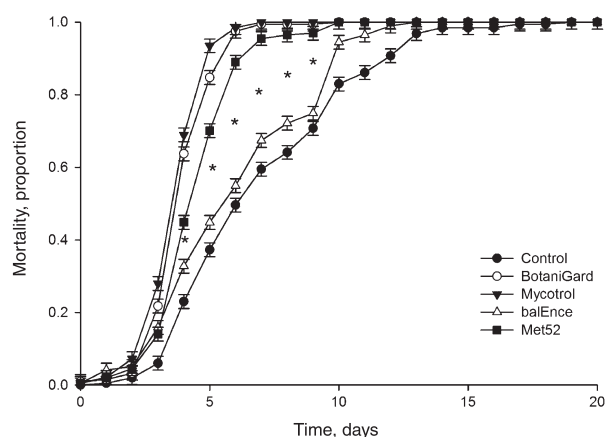


Fig. 1. Mean $\pm$ standard error mortality in adult house flies (*Musca domestica* L.; male and female combined) exposed to commercial fungal formulations on filter paper at a concentration of 1.57×10^6 conidia/cm² and containing three fungal strains. BotaniGard[®] ES and Mycotrol[®] O included *Beauveria bassiana* strain GHA, balEnce[™] contained *B. bassiana* strain HF23, and Met52[®] EC contained *Metarhizium brunneum* strain F52. All product formulations were emulsifiable suspensions using petroleum distillates, except Mycotrol[®] O, which used vegetable oils. Horticultural oil was incorporated as a control and used as a diluent. Asterisks indicate significant differences between BotaniGard[®] ES, Mycotrol[®] O and Met52[®] EC compared with balEnce[™] and the control (least significant differences test with Bonferroni correction, $\alpha = 0.05$).

The rate of mortality was highest with Met52[®] EC, followed by Mycotrol[®] O, BotaniGard[®] ES and, finally, balEnce[™]. All stable flies were dead in the Mycotrol[®] O treatment by day 15, in the BotaniGard[®] ES treatment by day 16, and in the balEnce[™] treatment by day 20. In the Met52[®] EC and control treatments, a few individuals survived to the end of the experiment (day 20). Confidence intervals for the LT₅₀ and LT₉₀ values in male and female stable flies typically overlapped and thus did not differ between treatments (Fig. 4). Stable fly LT₅₀ and LT₉₀ values showed less variability among treatments than in house flies, with overlapping confidence intervals between most treatments.

Sporulation

There was a significant effect of fungal treatment ($F_{4,12} = 3.90$, $P = 0.0296$) on sporulation in house flies (Fig. 5A). The highest rate of sporulation was found with Mycotrol[®] O, followed by Met52[®] EC, BotaniGard[®] ES and, finally, balEnce[™]. House flies treated with the fungal formulations BotaniGard[®] ES, Mycotrol[®] O and Met52[®] EC sporulated significantly more than those in the control treatment. At least 10% of house flies treated with these products sporulated. House flies treated with the fungal formulation balEnce[™] did not sporulate significantly more than those in the control treatment.

Similarly, there was a significant effect of fungal treatment ($F_{4,16} = 8.41$, $P = 0.0007$) on sporulation in stable flies (Fig. 5B). The rate of sporulation was highest with BotaniGard[®] ES, followed by Met52[®] EC, Mycotrol[®] O and, finally, balEnce[™]. All stable flies treated with fungal formulations sporulated

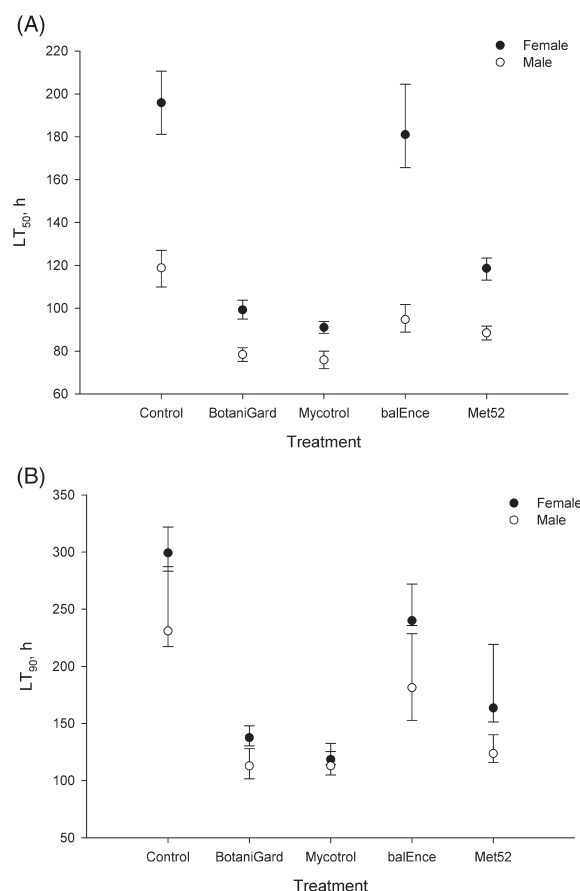


Fig. 2. Lethal time to (A) 50% (LT₅₀) and (B) 90% (LT₉₀) mortality in male and female adult house flies (*Musca domestica* L.) exposed to commercial fungal formulations on filter paper at a concentration of 1.57×10^6 conidia/cm² and containing three fungal strains. BotaniGard[®] ES and Mycotrol[®] O included *Beauveria bassiana* strain GHA, balEnce[™] contained *B. bassiana* strain HF23, and Met52[®] EC contained *Metarhizium brunneum* strain F52. All product formulations were emulsifiable suspensions using petroleum distillates, except Mycotrol[®] O, which used vegetable oils. Horticultural oil was incorporated as a control and used as a diluent. LT₅₀ and LT₉₀ $\pm$ 95% confidence intervals.

significantly more than flies in the control treatment. Stable flies in three of the fungal treatments, BotaniGard[®] ES, Mycotrol[®] O and Met52[®] EC, sporulated significantly more than stable flies in the balEnce[™] treatment. At least 15% of stable flies treated with these three products sporulated.

Discussion

In this study, three *B. bassiana* and one *M. brunneum* commercial formulation were tested against house flies and stable flies. The products BotaniGard[®] ES, Mycotrol[®] O and Met52[®] EC were the most effective in causing house fly and stable fly mortality and sporulation, whereas balEnce[™] was less effective. Faster rates of mortality were observed in house fly groups treated with BotaniGard[®] ES, Mycotrol[®] O and Met52[®] EC

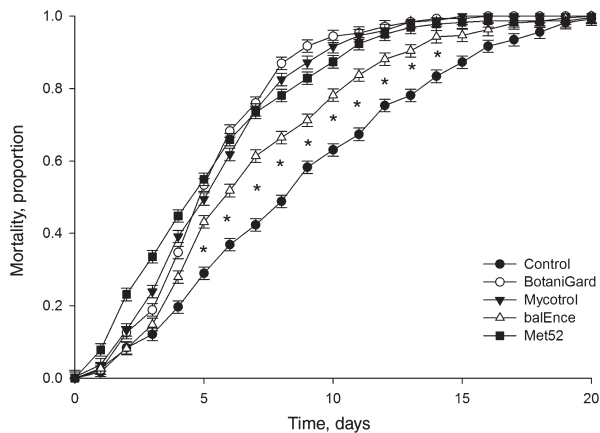


Fig. 3. Mean $\pm$ standard error mortality in adult stable flies [*Stomoxys calcitrans* (L.); male and female combined] exposed to commercial fungal formulations on filter paper at a concentration of 1.57×10^6 conidia/cm² containing three fungal strains. BotaniGard[®] ES and Mycotrol[®] O included *Beauveria bassiana* strain GHA, balEnce[™] contained *B. bassiana* strain HF23, and Met52[®] EC contained *Metarhizium brunneum* strain F52. All product formulations were emulsifiable suspensions using petroleum distillates, except Mycotrol[®] O, which used vegetable oils. Horticultural oil was incorporated as a control and used as a diluent. Asterisks indicate significant differences between BotaniGard[®] ES, Mycotrol[®] O, Met52[®] EC and balEnce[™] compared with the control (least significant differences test with Bonferroni correction, $\alpha = 0.05$).

(LT₅₀: < 120 h or 5 days) compared with those treated with balEnce[™] and the control treatment (LT₅₀: < 192 h or 8 days). Furthermore, in the house fly, balEnce[™] did not produce significantly higher sporulation than the control treatment. In the stable fly, all formulations were more effective in causing fly mortality and sporulation than the control treatment. House fly mortality was observed to be faster in males than females, possibly as a result of differences in body size or natural lifespan. Within fungal formulations, mortality rates in stable flies were slower in general with all treatments (LT₅₀: < 168 h or 7 days) compared with those in house flies. Differences in mortality and sporulation within and between fly species may result from strain origin, fungal species, exposure method (the bioassays were conducted with slight variations to account for factors specific to each species), dosage and fly species-specific physiology. To the present authors' knowledge, these formulations have not been evaluated for mortality and subsequent sporulation in house flies or stable flies using passive exposure methods prior to this study.

Formulation may inhibit or enhance the effectiveness of a fungus as a biopesticide. Proprietary formulations are likely to differ among the commercial products used in this study, although specific ingredients in the individual products were unknown. The components of these formulations, including surfactants and oils, may have contributed to their overall effectiveness in inducing mortality or increasing sporulation. The control substance used in this study was horticultural oil and although this may have induced some mortality, it was considered important in view of the use of oil-based fungal formulations. It is possible that a water-based control might have resulted in lower mortality, but its effects would not have been directly comparable with

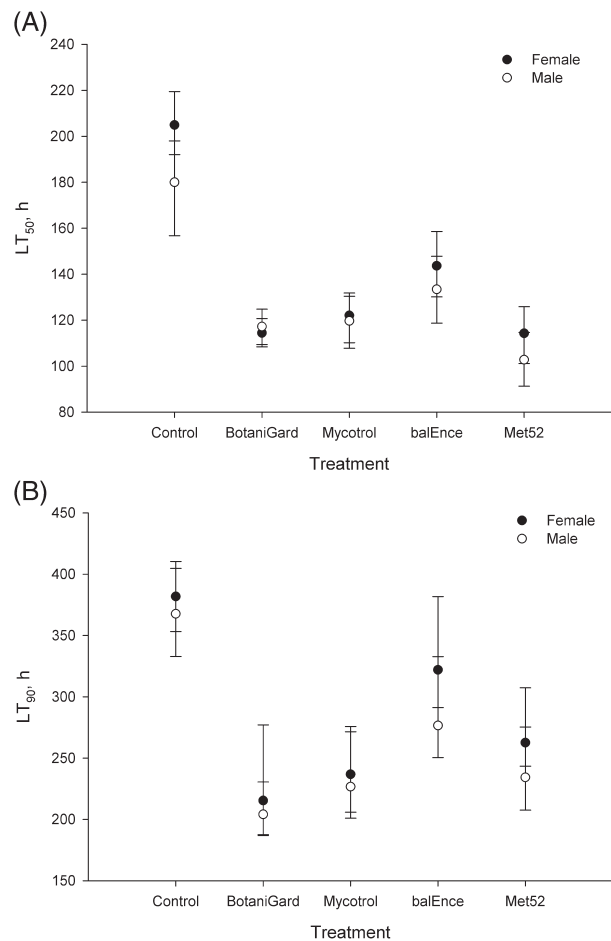


Fig. 4. Lethal time to (A) 50% (LT₅₀) and (B) 90% (LT₉₀) mortality in male and female adult stable flies [*Stomoxys calcitrans* (L.)] exposed to commercial fungal formulations on filter paper at a concentration of 1.57×10^6 conidia/cm² and containing three fungal strains. BotaniGard[®] ES and Mycotrol[®] O included *Beauveria bassiana* strain GHA, balEnce[™] contained *B. bassiana* strain HF23, and Met52[®] EC contained *Metarhizium brunneum* strain F52. All product formulations were emulsifiable suspensions using petroleum distillates, except Mycotrol[®] O, which used vegetable oils. Horticultural oil was incorporated as a control and used as a diluent. LT₅₀ and LT₉₀ $\pm$ 95% confidence intervals.

those of the formulations. It is known that surfactant compatibility can change the effectiveness of *B. bassiana* (Mishra *et al.*, 2013) and tests of the efficacy of *M. brunneum* in different vegetable oils showed that many oils increased the speed to kill of the fungus in house flies (Barson *et al.*, 1994). The inclusion of oils may have assisted in the infection process by disrupting the outer waxy layer of the exocuticle (Bateman *et al.*, 1993). Mycotrol[®] O was the only one of the products tested to contain vegetable oil as an emulsifier; however, experimental results with Mycotrol[®] O did not differ from those achieved with the product using the same strain of *B. bassiana* (i.e. GHA) but a different emulsifier, BotaniGard[®] ES, in sporulation or mortality. Application instructions for all tested commercial products indicate that the product should be mixed with water prior to

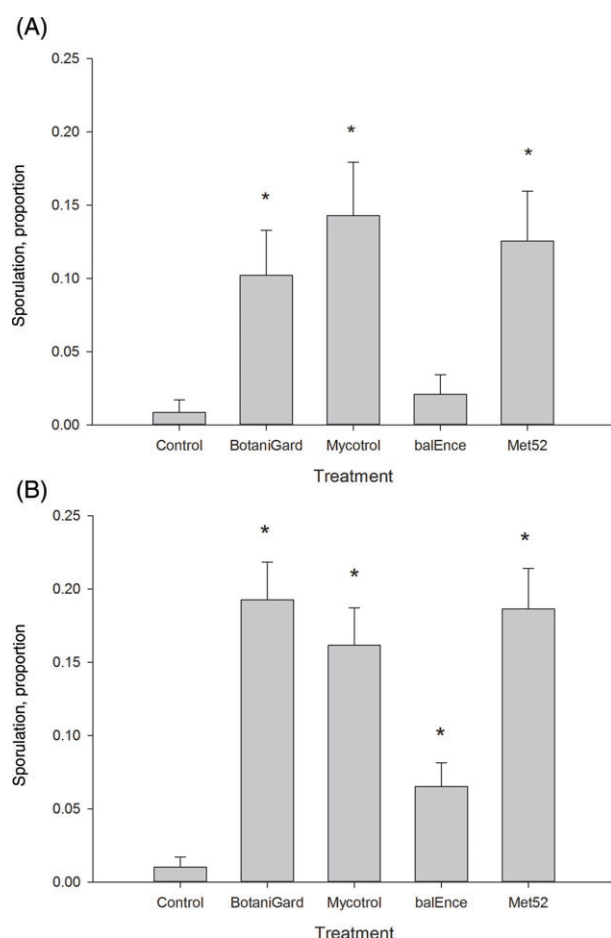


Fig. 5. Sporulation in adult (A) house flies (*Musca domestica* L.) and (B) stable flies [*Stomoxys calcitrans* (L.)] exposed to commercial fungal formulations on filter paper at a concentration of 1.57×10^6 conidia/cm² and containing three fungal strains. BotaniGard[®] ES and Mycotrol[®] O included *Beauveria bassiana* strain GHA, balEnce[™] contained *B. bassiana* strain HF23, and Met52[®] EC contained *Metarhizium brunneum* strain F52. All product formulations were emulsifiable suspensions using petroleum distillates, except Mycotrol[®] O, which used vegetable oils. Horticultural oil was incorporated as a control and used as a diluent. Asterisks indicate significant differences between fungal treatments and the control (least significant differences test, $\alpha = 0.05$).

application, and thus the most effective products in this study may include components that are more effective at protecting naturally hydrophobic conidia. Continuing work on the effects of formulation, combined effects of formulation and strain, and potential synergism to be elicited by combining fungal species (Mahmoud, 2009; Mwamburi *et al.*, 2009) may elucidate some of these differences and increase the effectiveness of commercial products by utilizing the additive effects of virulent strains and efficacious formulations.

Differences in the virulence of fungal strains may result from strain origin. Previous studies have found total adult fly mortality and time to death differed according to the strains of *B. bassiana* and *M. anisopliae* s.l. tested (Geden *et al.*, 1995; Watson *et al.*, 1995; Mishra *et al.*, 2011). Based on these issues, some authors have suggested that the search for new fungal strains should

concentrate on the pest species because specific strains may have higher virulence for particular hosts (Poprawski *et al.*, 1985; Steinkraus *et al.*, 1991). However, in the current study, the most effective strains were not originally collected from house flies or stable flies (GHA in BotaniGard[®] ES and Mycotrol[®] O, and F52 in Met52[®] EC), and the least effective product was formulated with a house fly-derived strain (HF23 in balEnce[™]). Interestingly, field trials demonstrated reduced fly numbers in poultry houses treated with balEnce[™] (Kaufman *et al.*, 2005). Perhaps even greater control would be realized in field situations if GHA or F52 strains were used. Samples of balEnce[™] used in the current study did not germinate in one preliminary trial, and had very low germination in another (<20%). In this case, product quality or consistency may be more of a factor than strain difference. It is also possible that the more septic environment in poultry houses in the study conducted by Kaufman *et al.* (2005), in comparison with the laboratory environment in the current study, promoted higher rates of mortality. This may also explain why balEnce[™] appeared to have an increased effect on stable flies in comparison with house flies. Microbial loads in stable fly containers were probably higher as a result of the presence of blood. Fungal species is also an important factor in fly mortality. In limited comparative studies, *M. anisopliae* s.l. has been shown to be more virulent than *B. bassiana* to adult house flies (Mishra *et al.*, 2011; López-Sánchez *et al.*, 2012), whereas stable fly susceptibility was dependent on fungal strain, but not necessarily fungal species (López-Sánchez *et al.*, 2012). However, it is important to note that the *M. anisopliae* lineage was revised in 2009 and a group of fungi formerly classified as *M. anisopliae* were reclassified as *M. brunneum* (Bischoff *et al.*, 2009). The strain F52 that is present in Met52[®] products represents one of the species found to be *M. brunneum*. The fungal strains tested by Mishra *et al.* (2011) and López-Sánchez *et al.* (2012) were not identified beyond the *M. anisopliae* species complex level, which is comprised of nine terminal taxa (Bischoff *et al.*, 2009). Therefore, the strains tested by earlier authors may belong to species that differ from that of the *M. brunneum* evaluated herein. In the current study, the mortality in both fly species induced by the *M. brunneum* product (Met52[®] EC) was similar to that induced by each of the two *B. bassiana* products (BotaniGard[®] ES and Mycotrol[®] O).

House fly and stable fly mortality response and sporulation differed by commercial product. House flies died sooner after fungal exposure than did stable flies. Other trials have found stable flies to be more resistant to *B. bassiana* (Watson *et al.*, 1995; López-Sánchez *et al.*, 2012), and it has been suggested that this difference may reflect cuticular or immunological differences between target species. The antimicrobial peptide stomoxin expressed in the foregut of adult stable flies is involved in epithelial immunity to microorganisms (Boulanger *et al.*, 2002). This peptide may be responsible for the delayed resistance to the fungal products. Alternatively, fly feeding behaviour may have contributed to faster mortality rates in house flies than stable flies. House flies frequently use their mouthparts on environmental surfaces and are likely to have had increased oral exposure to conidia during the experiment. Similarly, Mahmoud (2009) found increased effectiveness of tested *B. bassiana* and

M. anisopliae s.l. commercial products in the olive fly, *Bactrocera oleae* (Gmelin) (Diptera: Tephritidae) through oral exposure. Interestingly, greater sporulation was observed in stable flies than in house flies. As previously mentioned, the increased sporulation of stable flies may have resulted from the environment in which the flies were housed. The fungi may have benefited from the presence of blood and blood waste in the stable fly containers as additional growth media.

The effects of entomopathogenic fungi treatments are not as rapid as those of conventional pesticides. Such pathogens typically cause mortality at approximately day 5, which in this study was 3 days before mortality occurred in the control treatment. However, pre-lethal effects may enhance the impact on the fly population. A study in four poultry facilities recorded decreased adult house fly activity immediately after the initiation of treatment with balEnce™, followed by a decrease in house fly larvae 2 weeks later that was more dramatic and persistent than that observed with the comparative pyrethrin application (Kaufman *et al.*, 2005). These decreases in developing house fly abundance may be a result of the known pre-lethal effects of entomopathogenic fungi on fecundity, egg viability and larval development (Acharya *et al.*, 2015c; Machtinger *et al.*, 2016), combined with their effects on adult survival. Acharya *et al.* (2015c) estimated that treatment with entomopathogenic fungi resulted in a 90% reduction in fecundity and a 13–20% reduction in egg viability, which corresponded to a 21–26-fold decrease in reproductive rate. In another study, treated surfaces were found to decrease oviposition in house flies by more than 50% (Machtinger *et al.*, 2016). Therefore, despite a slow rate of kill, the impact of an entomopathogenic fungi treatment may be more persistent than that of a conventional pesticide application and thereby result in longer-term control.

The protection of livestock and their caretakers from the risks associated with high fly numbers can be increased with an integrated programme for fly management that includes biological control agents. *Beauveria bassiana* and *M. anisopliae* s.l. have been found to be compatible with natural parasitoids and predators of filth flies (Geden *et al.*, 1995; Kaufman *et al.*, 2005; Nielsen *et al.*, 2005), and may represent important tools in the management of fly pests without residual premise insecticides. Several of the commercial formulations tested demonstrated high pathogenicity to both house flies and stable flies. Future research on these strains and formulations may identify mechanisms to further increase efficacy.

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